Portal Uprights Research:

What are Portal Uprights:

A portal upright is an upright assembly that vertically offsets the wheel center below the CV axle centerline by integrating a gearbox or gear reduction system into the uprights. This increases ground clearance by allowing the drivetrain and suspension components to sit higher compared to the wheel center. Unlike traditional portal axles, which are typically integrated into solid axle assemblies, portal uprights are used in independent suspension assemblies.

(2) What will Portal Uprights Change:

(2.1) Geometric Changes:

- The wheel center drops below the CV line
- CV shaft sits higher
- Ground clearance increases under the drivetrain
- Upright height increases

(2.2) Drivetrain Changes:

- Gear train now at wheel
- Torque path is now changed: CV -> Gears -> Hub
- Possible gear reduction

(2.3) Suspension Effects:

- Higher Unsprung mass
- More mass outside chassis

(2.4) Performance Effects:

- Better ground clearance
- Lower CV angles
- Worse bump response
- Torque could be better or worse depending on gearing

Cornell's Findings:

- *Benefits:*

My Notes + Conclusions
Can use smaller tires
Added ground clearance
Potentially other suspension/steering benefits
I recommend using 2 gears
Most car companies use 4 gears
2 gears will help reduce weight, but have to run CV backwards to go forward
Try to shrink CV and use as light weight material as possible since stress is reduced on them to help offset the weight increase in uprights
WE NEED TO DECREASE OVERALL CAR WEIGHT lower car weight = less

Portal Uprights Research:

2 Gear:

- Simplest design
- Output is reversed unless drive train is modified
- Fewest amount of gears, shafts, and bearings
- Compact housing
- Can't really move output location, no to low amount of positioning freedom

  Summary: Simplest most efficient design also most lightweight, BUT we do have reverse drive shaft direction or flip drive train layout

3 Gear:

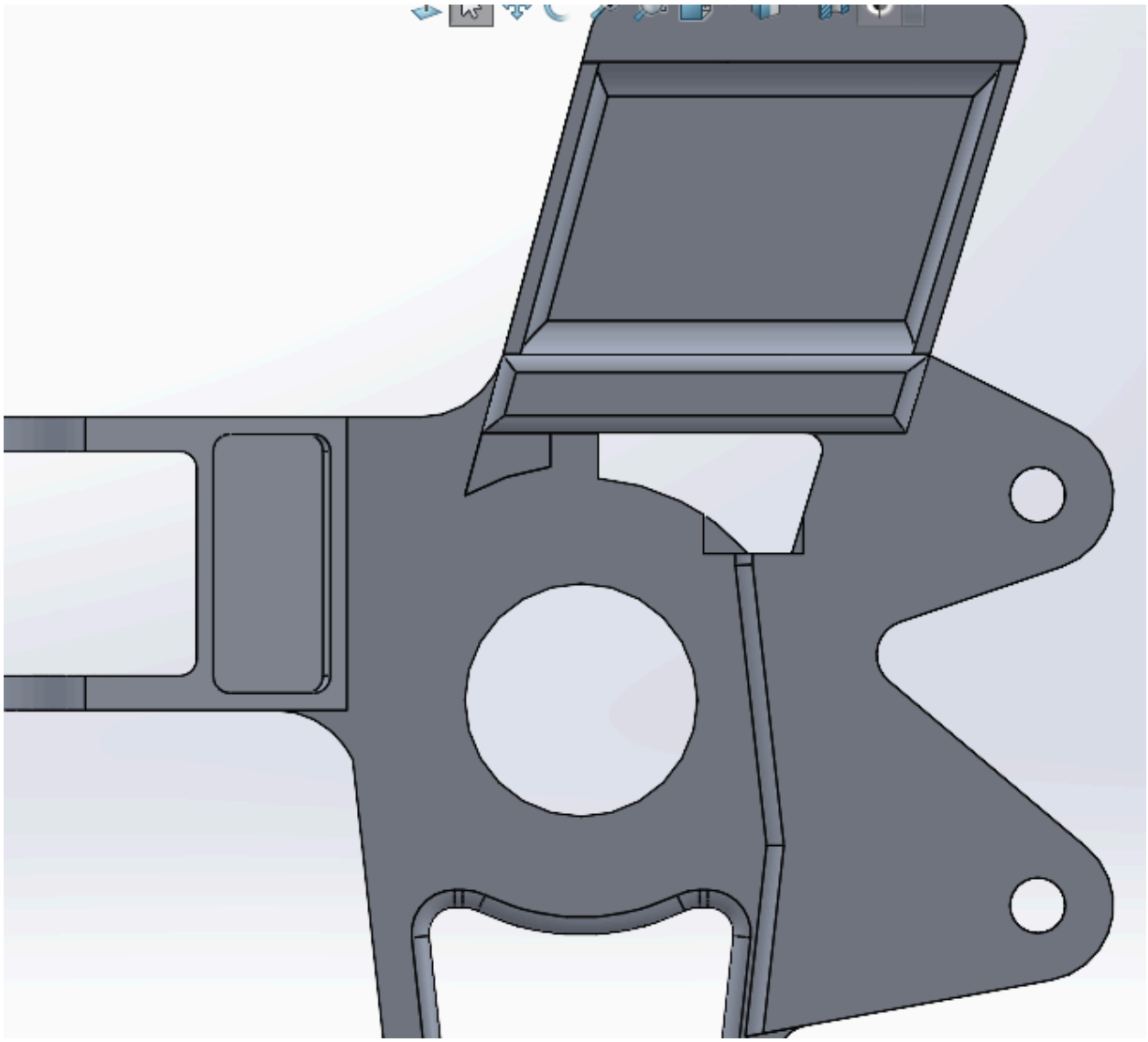
- Middle gear is an idler gear
- Idler reverses direction of output
- Can work with existing driveshaft
- Can move output location more freely
- Heavier than 2 gear: extra shaft, gear, and bearing
- Higher friction + larger housing

4 Gears:

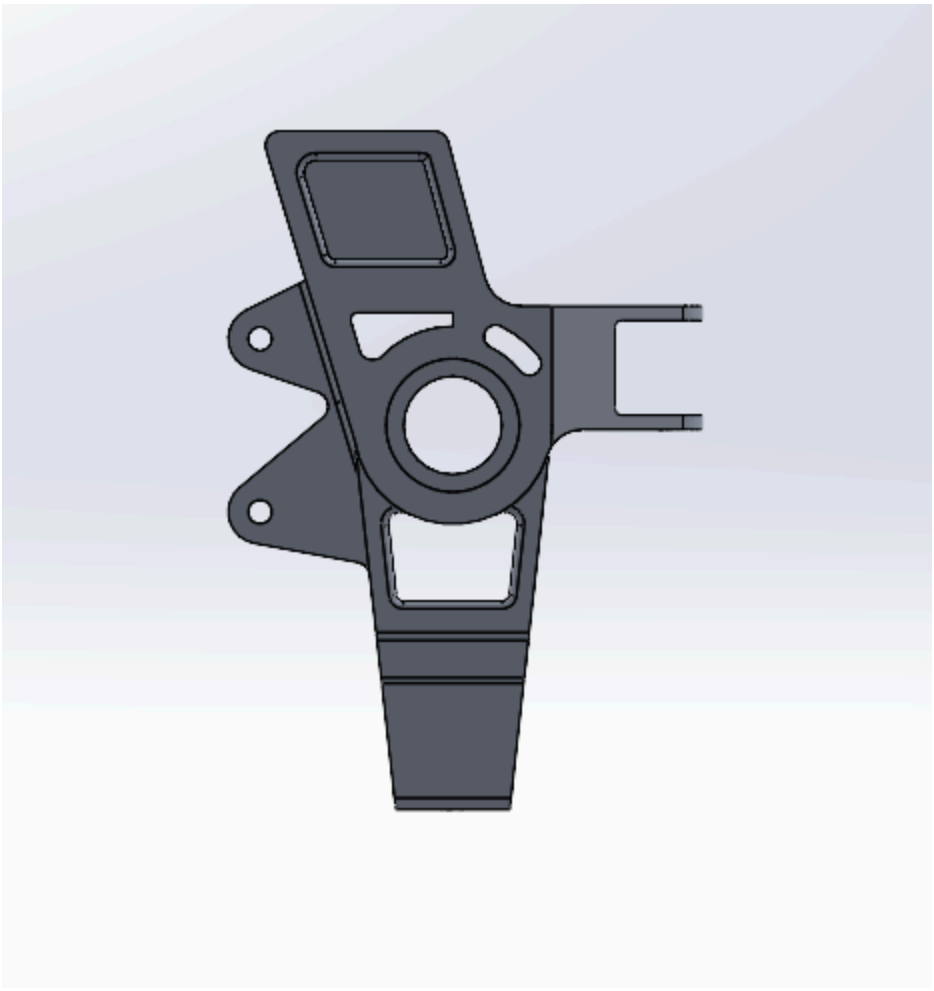
- Can place shafts where you'd like
- Could potentially help suspension? (Helped cornell with steering, suspension, CV axle alignment)
- Heaviest Option: Most amount of bearings, shafts, and gears
- Largest housing + most amount of frictional losses
- Hardest to manufacture + highly complex + Most expensive

  Summary: Highly recommended not going after this option Cornell decided not to use this, could be a cool future project after creating a simpler one. Cornell's was 6.9 lbs each portal upright (without oil) compared to their current 0.83 lbs upright. Ours is 1.84 lbs so I think going after a 4 gear portal upright is too ambitious for us right now

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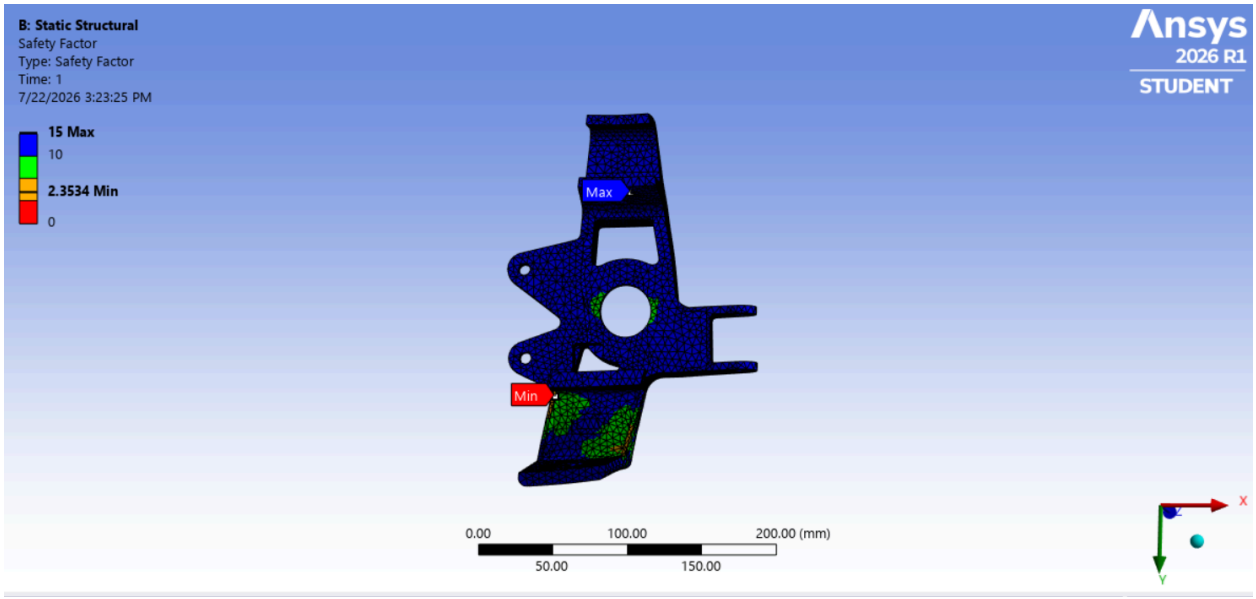


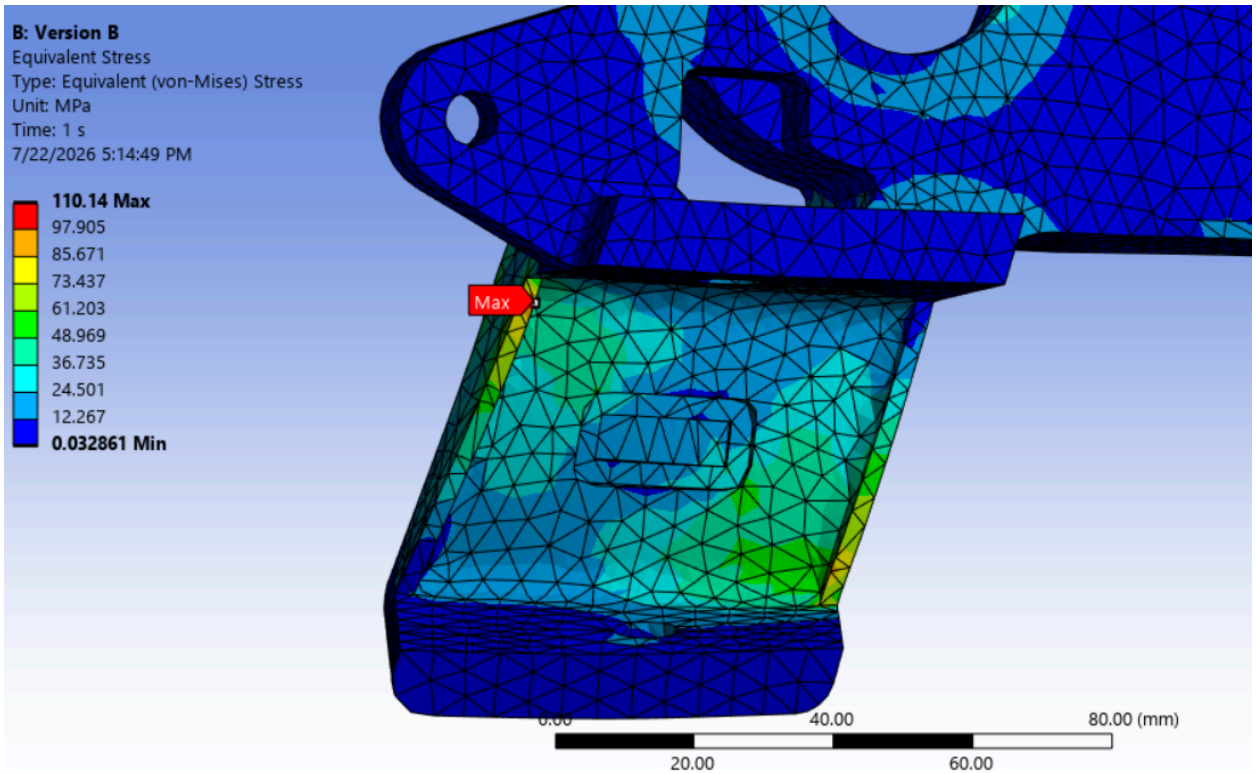
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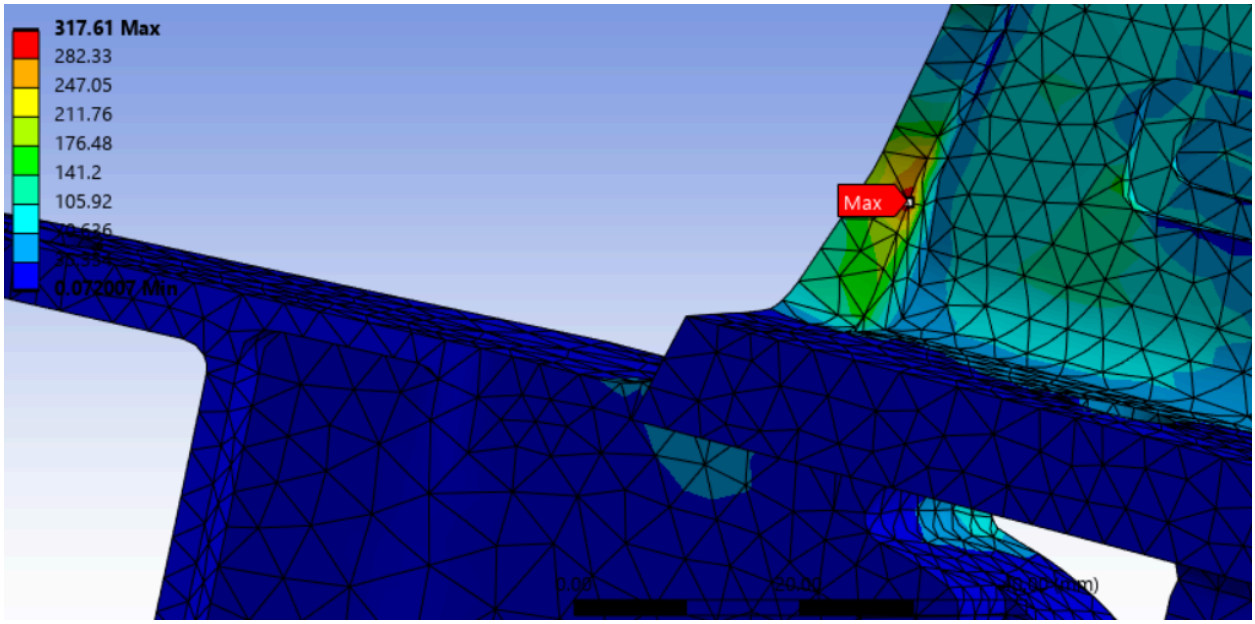
Version B of the Uprights:

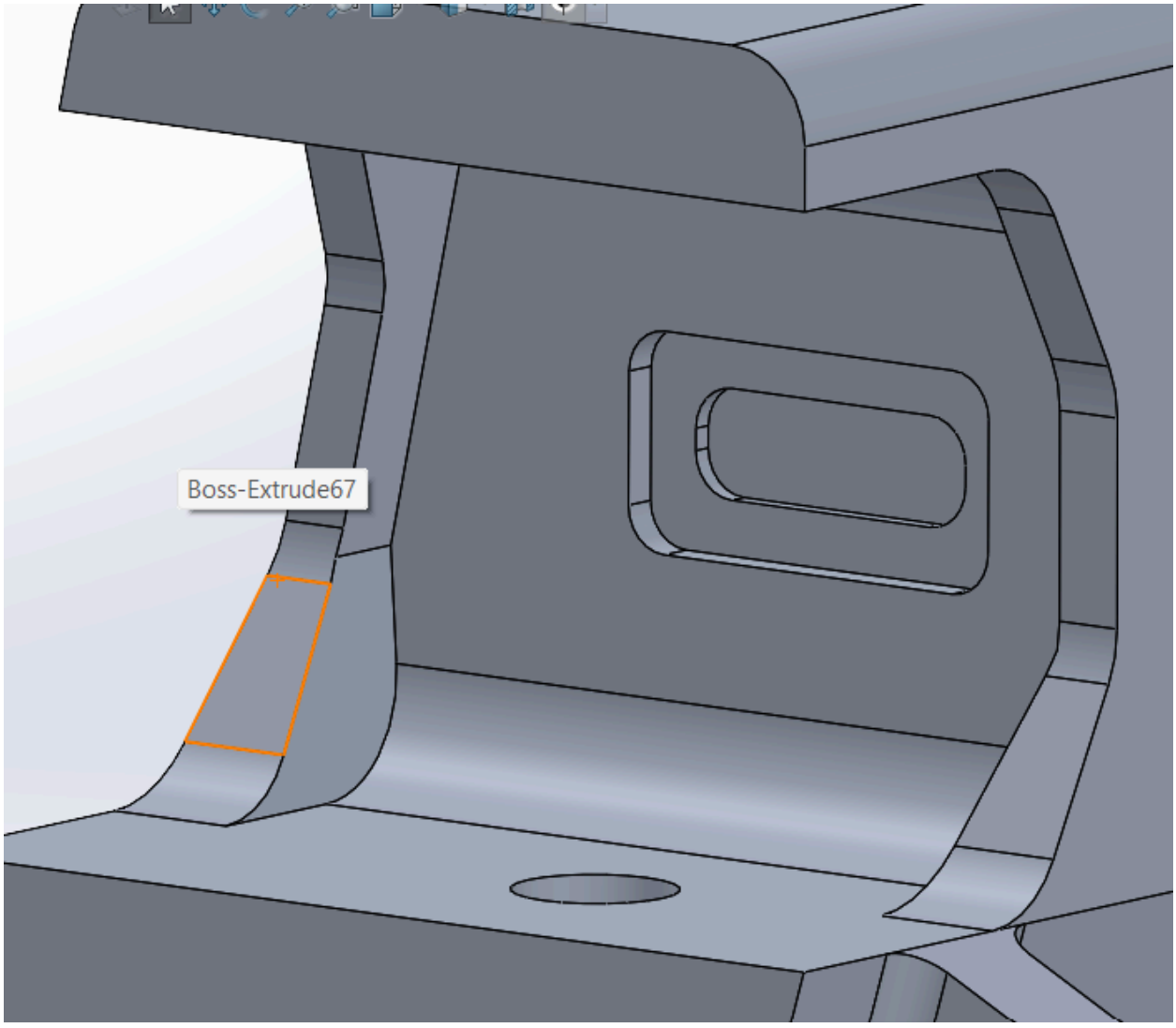
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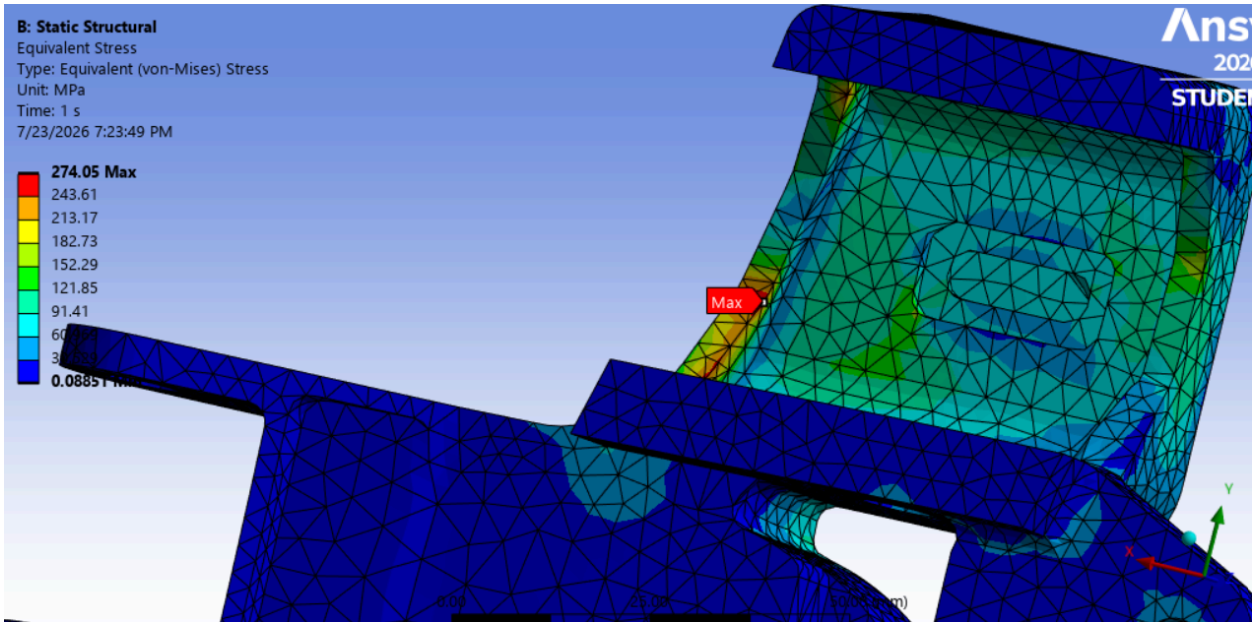


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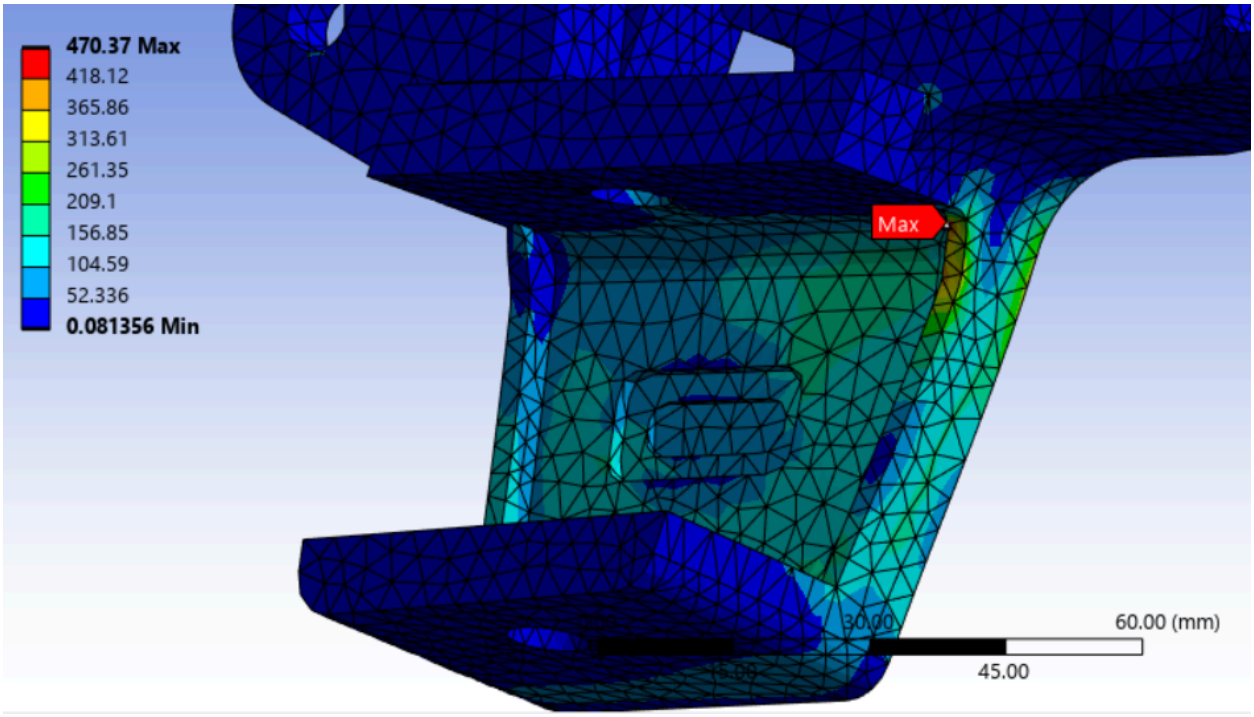




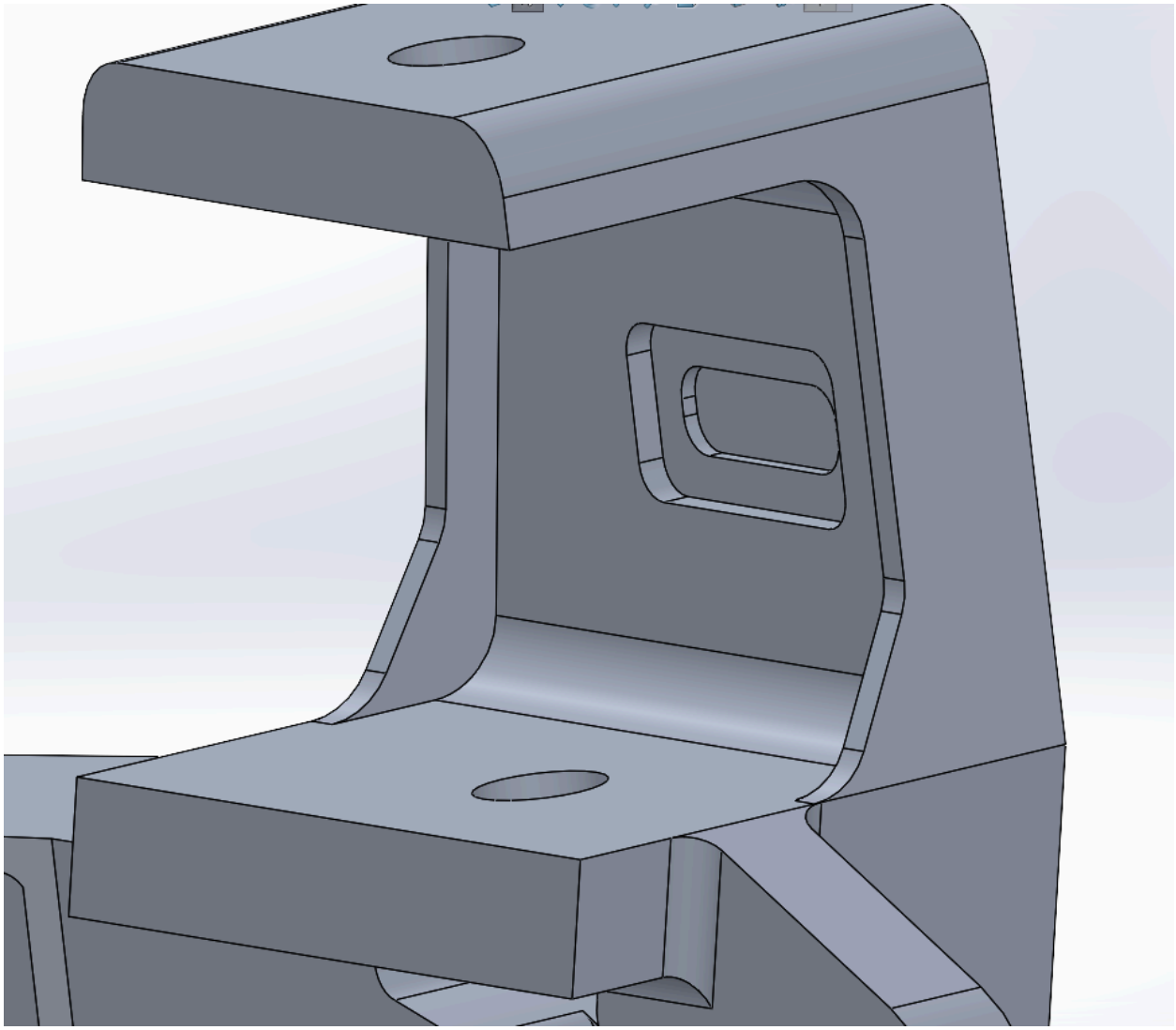
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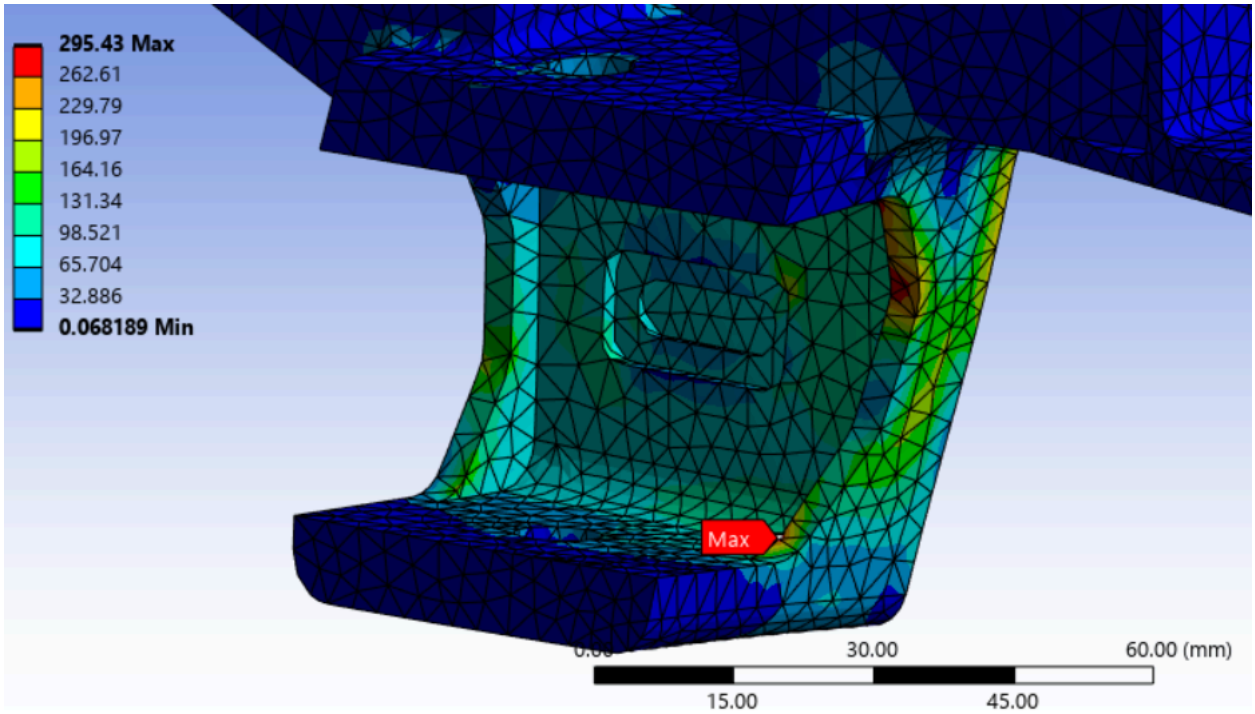
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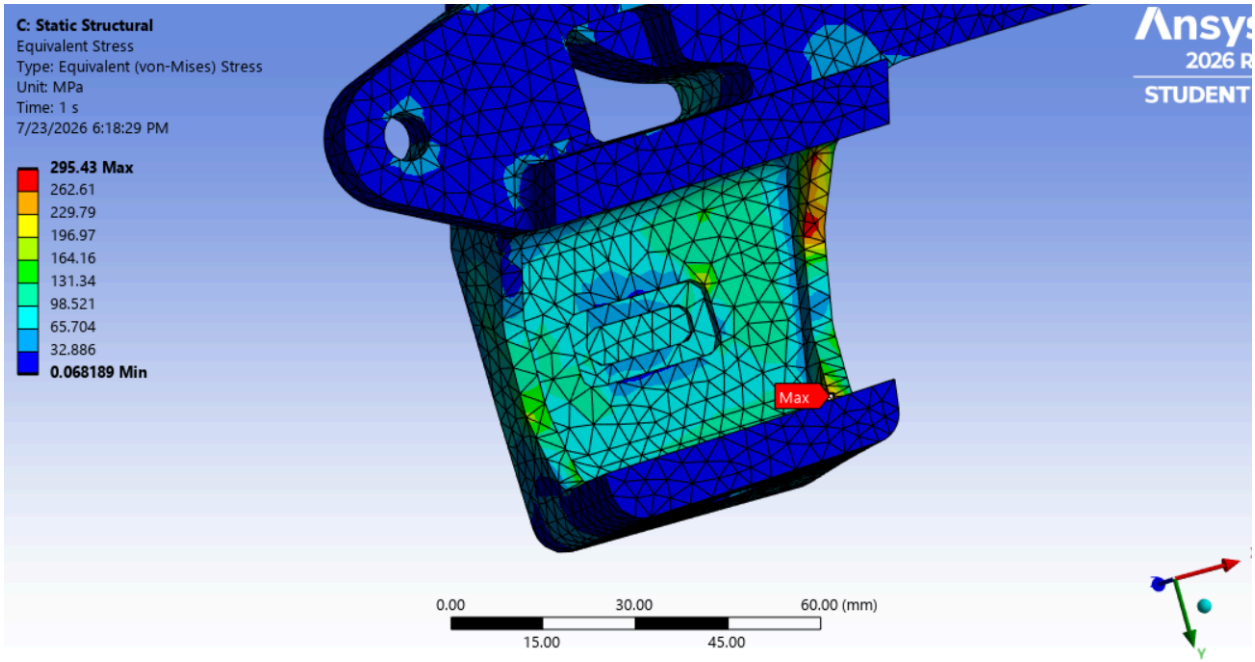


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